Dendroecology

Lesson 6 · 90 minutes · Reading forest history through tree growth

Learning objectives

Explain the questions addressed by dendroecology.

Relate growth changes, scars and ages to forest history.

Distinguish observed changes from hypotheses about causes.

Can trees preserve evidence of insect feeding?

What changes inside a stem when insects eat its leaves?

Separate a dated change from an inference about its cause.

Investigate forest history using ring widths, anatomy and comparisons.

R42 Khasanov & Sandlersky (2018)

Three kinds of evidence for forest history

Growth: changes in ring width and their duration.

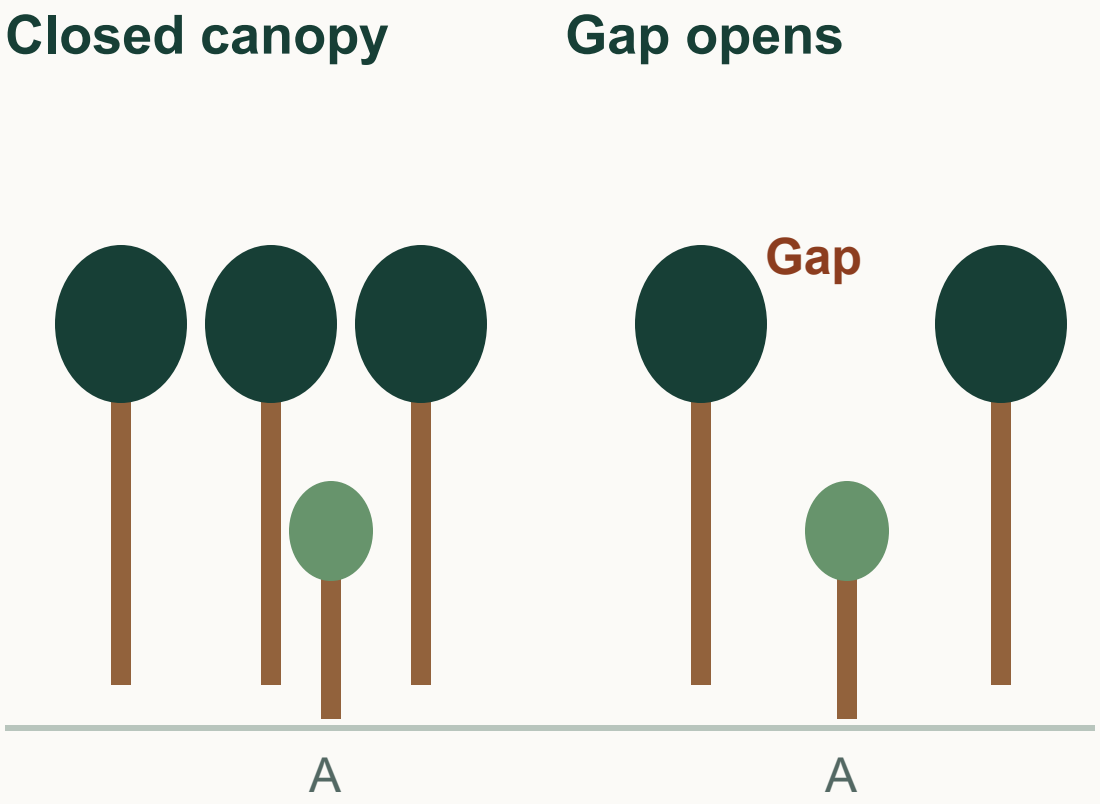
Injuries: dated scars and subsequent wound closure.

Populations: regeneration, establishment and mortality.

R35 Nowacki & Abrams (1997)

R40 Fastie et al. (2002)

Canopy gaps and changing competition



A canopy gap changes light and competitive conditions.

Responses depend on species, position and prior growth.

Reconstruct stand dynamics from multiple individual histories.

Schematic / synthetic teaching data, not empirical measurements or photographs.

R35 Nowacki & Abrams (1997)

Growth suppression and release

Suppression: a period of relatively slow growth.

Release: a sustained increase in growth to investigate.

One wide ring alone does not establish canopy opening.

R35 Nowacki & Abrams (1997)

R36 Black & Abrams (2003)

Different causes can produce similar changes

Consider competition, climate, water, insects and management.

Distinguish immediate changes from delayed responses.

Test a hypothesis with independent observations or records.

Plan the comparisons before sampling

Consider comparison stands or non-host species.

Record site, species, size, crown position and management.

Avoid selecting only affected trees or present-day survivors.

R39 Hirano & Morimoto (1999)

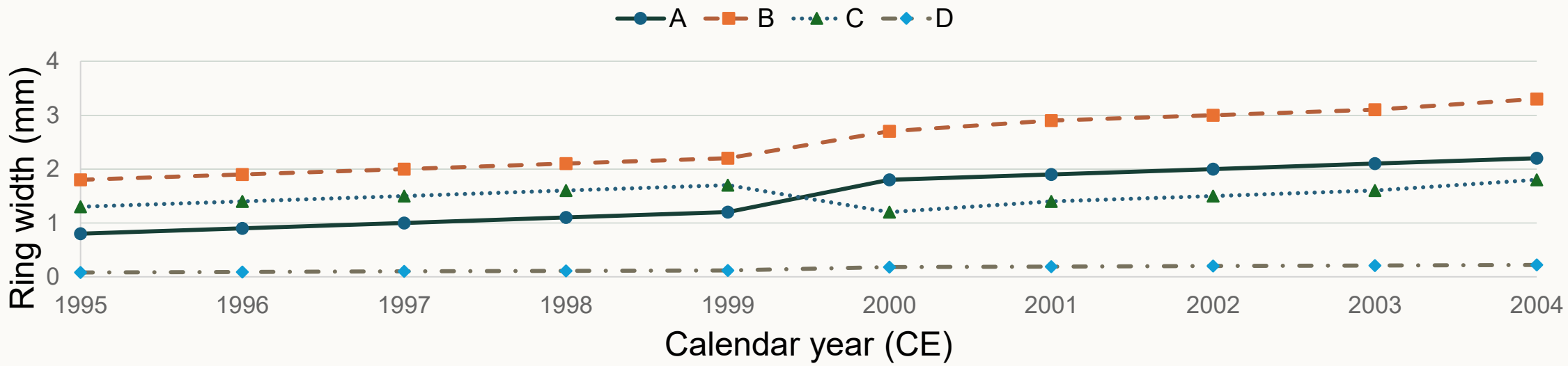
R40 Fastie et al. (2002)

Compare individual growth histories

A, B and D represent a target stand; C a comparison stand.

Which trees change around 2000?

Synchrony is evidence to investigate, not proof of a cause.



Schematic / synthetic teaching data, not empirical measurements or photographs.

R35 Nowacki & Abrams (1997)

Compare means before and after a change

M1: mean for 1995–1999; M2: mean for 2000–2004.

Percent change G = (M2 − M1) / M1 × 100%.

Five-year windows are a teaching choice, not a universal criterion.

Exercise A: does 100% mean the same thing?

Tree	M1 (mm)	M2 (mm)	Change G
A	1	2	Calculate
B	2	3	Calculate
C	1.5	1.5	Calculate
D	0.1	0.2	Calculate

Compare A and D using percentages and absolute changes.

Schematic / synthetic teaching data, not empirical measurements or photographs.

R35 Nowacki & Abrams (1997)

Exercise A: percentages and absolute changes

Tree	Change (%)	Difference (mm)
A	100%	1
B	50%	1
C	0%	0
D	100%	0.1

A and D both increase 100%, but by 1.0 and 0.1 mm.

Schematic / synthetic teaching data, not empirical measurements or photographs.

R35 Nowacki & Abrams (1997)

Release criteria depend on the trees

Small prior growth can produce large percentage changes.

Choose criteria appropriate to species and growth conditions.

Do not turn 100% into a universal disturbance threshold.

R36 Black & Abrams (2003)

Ring width and basal area increment

BAI measures the increase in stem cross-sectional area.

For a circular stem: $BAI = \pi[(r + w)^2 - r^2]$.

Previous radius r	Ring width w	Approximate BAI
10 mm	1 mm	66 mm²
30 mm	1 mm	192 mm²

Equal widths can add different areas; BAI alone is not biomass.

Schematic / synthetic teaching data, not empirical measurements or photographs.

R35 Nowacki & Abrams (1997)

Inferring establishment from tree ages

Establishment differs from the first ring at sampling height.

Consider missing pith and years needed to reach that height.

Present-day age structure does not recover every past individual.

R35 Nowacki & Abrams (1997)

The bias in looking only at survivors

Death, logging and decay remove older evidence.

Consider dead wood, stumps, regeneration and forest records.

Report sampling criteria and the number of trees through time.

R40 Fastie et al. (2002)

Reading fire history from tree scars

Date fire scars and assess the nature of the injury.

Giant-sequoia studies combine scars and ages to study fire history.

A fire may leave no scar: no scar does not prove no fire.

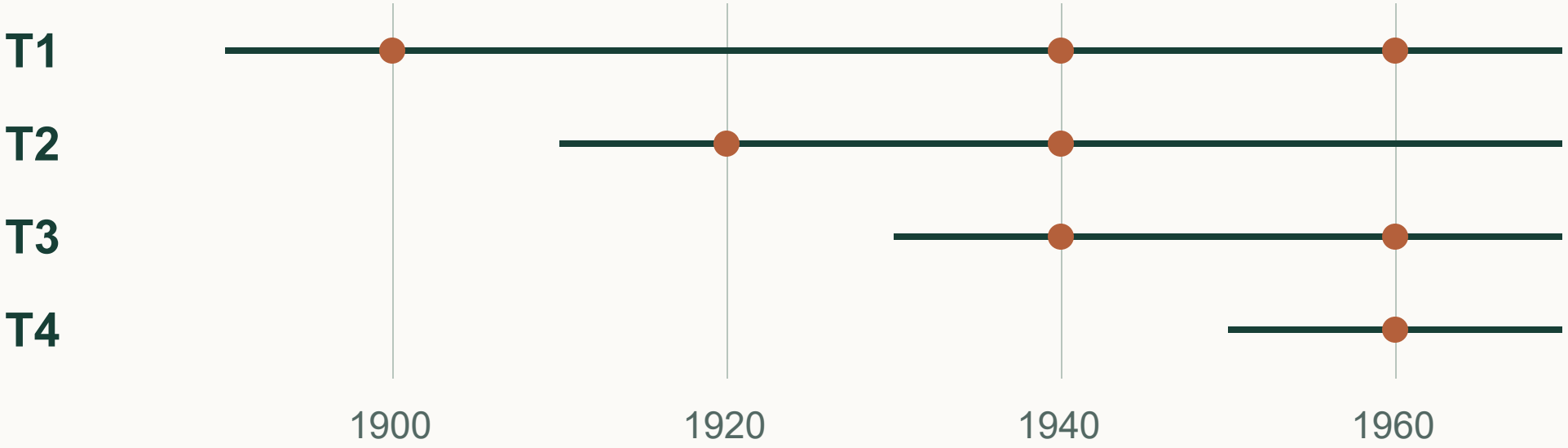
R38 US National Park Service

R40 Fastie et al. (2002)

Recording periods differ among trees

Show recording periods alongside scar dates.

Separate not recording from recording without a scar.



Schematic / synthetic teaching data, not empirical measurements or photographs.

R37 NOAA NCEI

R40 Fastie et al. (2002)

Exercise B: count scarred and recording trees

	1900	1920	1940	1960
T1	1	0	1	1
T2	—	1	1	0
T3	—	—	1	1
T4	—	—	—	1

1 = scar; 0 = recording, no scar; — = not recording

Count recording trees and scarred trees for each year.

Schematic / synthetic teaching data, not empirical measurements or photographs.

R37 NOAA NCEI

R40 Fastie et al. (2002)

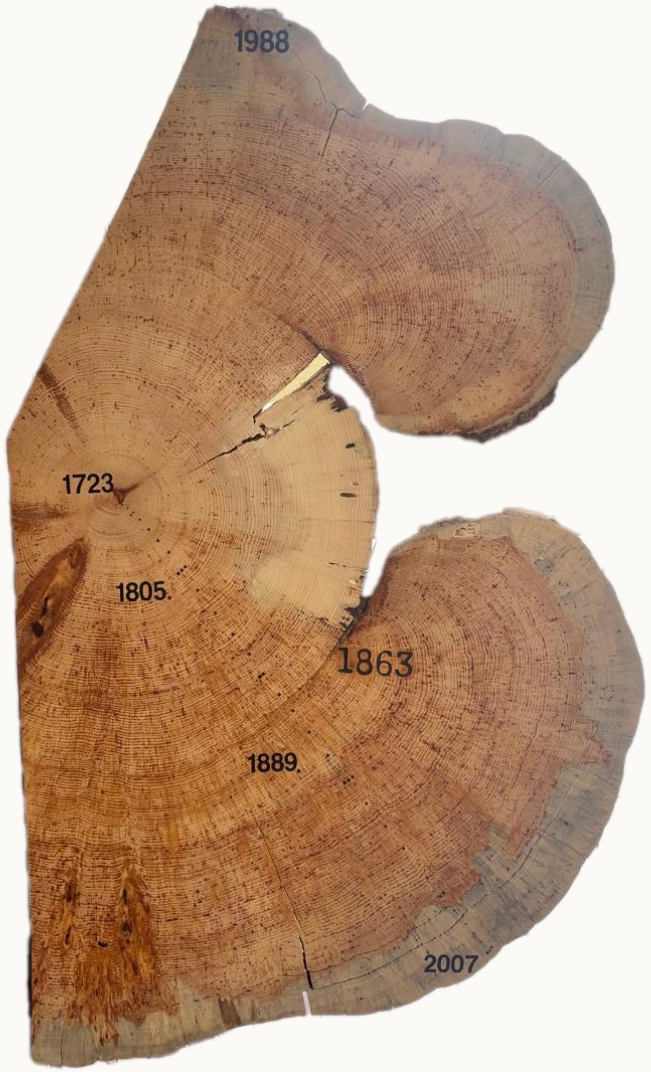
Exercise B: the denominator changes too

Year	Recording	Scarred	Percentage
1900	1	1	100%
1920	2	1	50%
1940	3	3	100%
1960	4	3	75%

These sample percentages are not burned area or fire severity.

Schematic / synthetic teaching data, not empirical measurements or photographs.

Fire and environmental history in one tree



Cross section of a whitebark pine

1723 and 1863: scars record fire damage

Growth after the 1863 fire formed wood on both sides of the large scar.

A scar records local injury to this tree. Compare other trees and independent records to examine the extent and cause of a fire.

Photo: USGS, “Cross-section of Whitebark Pine” / Public Domain

USGS: image and credit

USGS: specimen interpretation

Growth suppression caused by insects

Defoliation can alter growth and wood anatomy.

Compare host and non-host trees, damage records and anatomy.

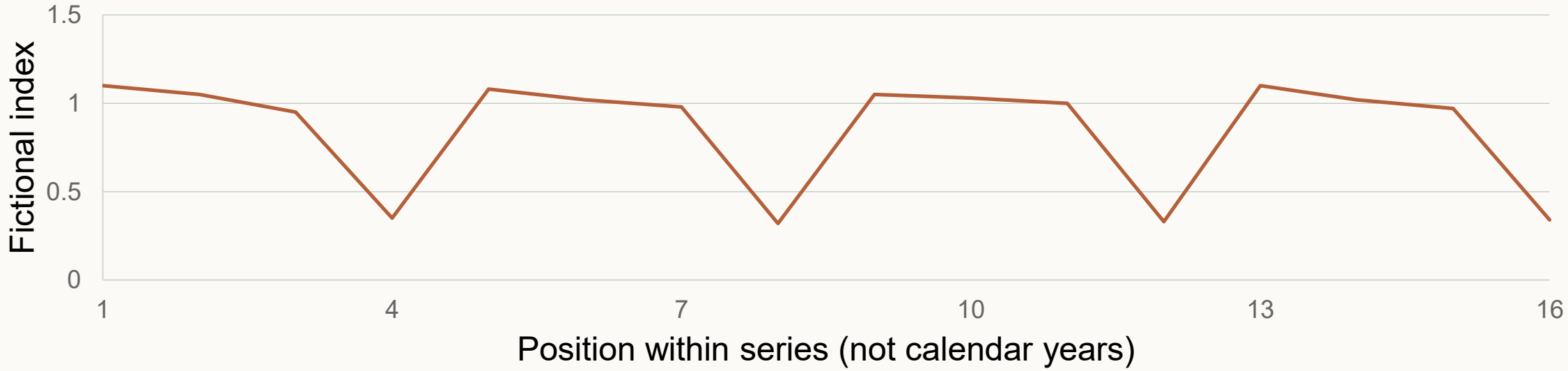
Recovery after insect damage need not indicate canopy opening.

What do periodic narrow rings indicate?

Cockchafer damage can produce periodic narrow oak rings.

Repetition can create apparent matches between different periods.

Check anatomy, damage records and independent dating evidence.



Schematic / synthetic teaching data, not empirical measurements or photographs.

R41 Dendrochronologia (2013)

Can insect damage be seen beyond ring width?

Khasanov & Sandlersky (2018) · Central Russia, 1878–2015

Oak (Quercus robur) **49 trees, three sites**	**Ash (Fraxinus excelsior)** **21 trees at one of those sites**
Ring width + wood anatomy	Non-host comparison
Intra-annual density fluctuations (IADFs)	
Abrupt growth changes (AGC)	

An IADF alone does not diagnose insect damage: compare evidence.

Summary of the study design; source anatomical images and data plots are not reproduced.

R42 Khasanov & Sandlersky (2018)

How did surviving infested trees respond?

Growth: six sites. Wood anatomy: two trees at one site.

Three of six sites declined; three showed no change or an increase.

Do survivors represent the full population, including trees that died?

R44 Virginia Tech (2014)

R45 Walker et al. (2014)

A Japanese study: black pine and air pollution

Hirano and Morimoto (1999) compared polluted and unpolluted stands.

They assessed reduced growth alongside SO_2 and a climate model.

Read how comparisons and other evidence support the interpretation.

Reading eruptions: Mount St. Helens

The independently documented eruption occurred on 18 May 1980.

Damage and survival vary across the landscape.

Compare rings, scars and establishment with deposits and terrain.

Separate the disturbance year from the growth-response year.



Photo: Lyn Topinka / USGS, 22 August 1980. Elk Rock area.

USGS: photograph and credit

US Forest Service: forest research

Reading earthquakes: Wrightwood

Separate the event year from the year of a tree-growth response.

December 1812

Historical earthquake record

→

From 1813

Sustained growth reduction in some trees near the fault

Drought can also produce narrow rings → compare trees away from the fault.

Question: what independent evidence distinguishes earthquake damage from drought?

Conceptual summary of the published interpretation; not a measured ring-width graph.

Jacoby et al. (1988): paper

Caltech: earthquake record

Compare hypotheses using different evidence

Release plus logging records: investigate thinning or removal.

Host-specific suppression plus records: investigate insect damage.

Scars, recruitment and archives: investigate postfire development.

R39 Hirano & Morimoto (1999)

R40 Fastie et al. (2002)

Choose processing to preserve the question

Persistent changes may be the signal of interest.

Check whether detrending removes the change being studied.

Keep raw series, processing choices and sensitivity checks.

R35 Nowacki & Abrams (1997)

R36 Black & Abrams (2003)

From a question to a plan or review

Plan: define the system, hypothesis and expected evidence.

Review: select studies and compare methods and findings.

For either, consider alternative explanations and limits.

R35 Nowacki & Abrams (1997)

Exit questions

Does a 100% growth increase prove a canopy gap?

How do you treat no-scar years and non-recording periods?

Does a first ring dated 1980 at sampling height prove germination in 1980?

R35 Nowacki & Abrams (1997)

Next: extracting climate signals

Lesson 7 introduces dendroclimatology.

Examine species, site and seasonal climate responses.

Keep competition, disturbance and insects in view as other influences.

R35 Nowacki & Abrams (1997)